COAL - Assignment 2

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Q.1

```
mov  ax, 3
mov  bx, ax
mov  cx, ax

shl  ax, 1
shl  bx, 3
shl  cx, 4
add  ax, bx
add  ax, cx
```

26 = 16 + 8 + 2

Q.2

```
mov  eax, 02345ABCDh
mov  edx, 0
mov  ecx, 32
   L1:
      shl  eax, 1
      rcl  edx, 1
      loop L1

mov  eax, 02345ABCDh
mov  edx, 0
mov  ecx, 32
      L2:
       shr eax, 1
       rcr edx, 1
       loop L2
```

Made with Goodnotes

Q.3
```
mov   eax, 02345ABCDh
mov   edx, 0
mov   ecx, 32
   L1:
     shl  eax, 1
     rcl  edx, 1
     loop L1
```

```
mov   eax, 02345ABCDh
mov   edx, 0
mov   ecx, 32
     L2:
      shr eax, 1
      rcr edx, 1
      loop L2
```

Q.5

dx = 9A3Fh

ax = 7F80h

CF = 0

ax = 106Ah ∴ 7F80 + 90EA CF = 1

dx = 9A41h ∴ 9A3F + 1 + 1 CF = 0

CF = 1

ax = 918Ah ∴ CF = 0 106A + 8120

dx = 9A40h ∴ 9A41 + 0FFFF CF = 1

ax = 918Ah dx = 9A40h CF = 1

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4. INCLUDE Irvine32.inc

```
.data
    BSeconds  BYTE ?
    BMinutes  BYTE ?

.code
    main PROC

        mov dx, 0101100111011010b
        mov ax, dx
        and al, 00011111b
        mov BSeconds, al

        mov  ax, dx
        shr  ax, 5
        and  al, 00111111b
        mov  BMinutes, al

        movzx eax, BSeconds
        call  WriteDec
        call  Crlf
        movzx eax, BMinutes
        call  WriteDec
        call  Crlf

          exit
    main  ENDP
    END   main
```

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Q 6

a) test ax, ax
b) test bx, 1
c) test dx, dx
d) test dx, 8000h
e) test bL, bL

Q.7

dx = 0005h
ax = 0010 h
CF = 1
dx = 0016 h ∴ 0005 + 0010 + 1

Q.8

1.	mov ecx, 8	we need ecx, other wise garbage value
2.	mov esi, OFFSET val1 mov edi, OFFSET val2	assign address not value
3.	mov ebx, OFFSET result	to seb answer to result
4.	mov BYTE PTR [ebx], al	over-writes val1, need ans in result
5.	inc esi inc edi	Little-endian so we go from lowest to highest byte
6.	inc ebx	increment for result too

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9.
```
INCLUDE Irvine32.inc
.data
    msg1    BYTE    "enter your marks:", 0
    msg2    BYTE    "your grade is:", 0
    grade   BYTE    ?

.code
    main PROC
    mov edx, OFFSET msg1
    call WriteString
    call ReadInt

    cmp eax, 80
    jge A
    cmp eax, 60
    jge B
    cmp eax, 40
    jge C
    jmp F

A:
    mov grade, 'A'
    jmp print
B:
    mov grade, 'B'
    jmp print
C:
    mov grade, 'C'
    jmp print

F:
    mov grade, 'F'
    jmp print

print:
    mov edx, OFFSET msg2
    call crlf
    call WriteString
    mov al, grade
    call WriteChar

    exit
    main ENDP
    END main
```

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Q. 10

eax = 3

ebx = 7

[3]

[3, 7]

eax = 7 , [3]

[3, 10]

ebx = 10 , [3]

eax = 3

eax = 3	ebx = 10

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11.

```
INCLUDE Irvine32.inc
.data
    msg1    BYTE    "enter number:", 0
    msg2    BYTE    "result is:", 0
    num1    SDWORD  ?
    num2    SDWORD  ?
    num3    SDWORD  ?
    num4    SDWORD  ?
    result  SDWORD  ?
.code
    main PROC
    mov edx, OFFSET msg1
    call WriteString
    call ReadInt
    mov num1, eax

    mov edx. OFFSET msg1
    call WriteString
    call ReadInt
    mov num2,eax

    mov edx, OFFSET msg1
    call WriteString
    call ReadInt
    mov num3, eax

    mov edx, OFFSET msg1
    call WriteString
    call ReadInt
    mov num4, eax

    mov eax, num1
    cmp eax, num2
    jg secondPartOfIF
    jmp Elself

secondPartofIf:
    mov eax, num3
    cmp eax, num4
    je Elself
    mov ebx, num1
    add ebx, num3
    mov result, ebx
    jmp print

Elself:
    mov eax, num2
    cmp eax, num4
    jne Else
    mov ebx, num2
    sub ebx, num1
    mov result, ebx
    jmp print

Else:
    mov result,0
    jmp print
```

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```
print:
    call crlf
    mov edx, OFFSET msg2
    call WriteString
    mov eax, result
    call WriteInt
    call crlf

exit
    main ENDP
    END main
```

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12.
```
INCLUDE Irvine32.inc
.data
op1     SDWORD   15
op2     SDWORD   30
x       SDWORD   8
y       SDWORD   12
z       SDWORD   0

.code
main PROC

while1:
    mov  eax, op1
    cmp  eax, op2
    jle  if1
    jmp  endWhile

if1:
    mov  eax, op1
    cmp  eax, x
    jge  if2
    jmp  elseBlock

if2:
    mov  eax, op1
    cmp  eax, y
    jle  R1
    jmp  elseBlock

elseBlock:
    sub  z, 10
    jmp  decrem

R1:
    add  z, 10
    jmp  decrem

decrem:
    dec  op1
    jmp  while1

endWhile:
    mov  eax, z
    call WriteInt
    call Crlf
    exit
main ENDP
END main
```

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